INTRODUCTION TO GRAPH THEORY · PROBLEM SET

Degrees, Regularity, Isomorphism and Connectivity

Complete worked solutions to eight problems

Eight problems · Full proofs and constructions · All results machine-verified

This document contains complete solutions to the eight problems of the set. Every claim is proved from first principles: the only facts assumed are the definitions collected in the section below. Wherever a problem asks for a construction (a graph with a prescribed degree sequence, an explicit isomorphism, a counterexample), the object is exhibited and then verified entry by entry rather than merely asserted to exist.

Four of the problems have a purely computational shadow — the reachability question in Problem 1, the realisability questions in Problem 2, the isomorphism question in Problem 6, and the existence question in Problem 8. For each of these the human proof given here was additionally confirmed by exhaustive computer search (breadth-first search over all positions, brute-force search over all $9! = 362\,880$ vertex bijections, and a max-flow realisation of the prescribed degree pairs). Those checks are reported at the end of the relevant sections; they are corroboration, not substitutes for the arguments.

Every solution below is carried out by graph-theoretic means. Two of the problems are not posed in the language of graphs at all — Problem 1 is a chess puzzle, Problem 8 a counting riddle about a network — and in each the first move is a *reduction*: replace the object by a graph whose structure answers the question. The appendix on the final page records, problem by problem, which graph-theoretic objects the solution uses and which results are applied to them.

Contents

Notation, conventions, and three basic lemmas

Unless stated otherwise every graph below is *finite* and *simple*: finitely many vertices, no loops, no parallel edges. We write $G = (V(G), E(G))$, $n = |V(G)|$, $m = |E(G)|$. For a vertex v, $N_G(v) = \{u : uv \in E(G)\}$ is its *neighbourhood* and $d_G(v) = |N_G(v)|$ its *degree*; we drop the subscript when the graph is clear. As usual $\Delta(G) = \max_v d(v)$ and $\delta(G) = \min_v d(v)$, and G is *k-regular* if $d(v) = k$ for every vertex v. We write P_k, C_k, K_k for the path, the cycle and the complete graph on k vertices.

A *walk* of length k is a sequence $v_0 v_1 \dots v_k$ with $v_{i-1}v_i \in E(G)$ for all i; a *path* is a walk whose vertices are pairwise distinct; a *cycle* is a walk of length $k \ge 3$ with $v_0 = v_k$ and $v_0, \dots, v_{k-1}$ pairwise distinct. The *components* of G are its maximal connected subgraphs; G is *connected* if it has exactly one component. An isolated vertex is regarded as a trivial path $P_1 = K_1$. The *distance* $\operatorname{dist}(u, v)$ is the length of a shortest u–v path, and $\operatorname{diam}(G) = \max_{u,v} \operatorname{dist}(u, v)$.

The next three lemmas are used again and again, so we state them once here.

Lemma A (handshake lemma) *For every finite graph G we have $\sum_{v \in V(G)} d(v) = 2|E(G)|$.*

Proved in full in Problem 4, where it is the heart of the matter; we quote it freely before then.

Lemma B (every walk contains a path) *If G contains a u–v walk then G contains a u–v path.*

Proof. The set of lengths of u–v walks is a non-empty set of non-negative integers, so we may choose a u–v walk $W = v_0 v_1 \dots v_k$ ($v_0 = u$, $v_k = v$) of *minimum* length. Suppose some vertex occurred twice, say $v_i = v_j$ with $i < j$. Then $v_0 \dots v_i v_{j+1} \dots v_k$ is again a u–v walk, of length $k - (j - i) < k$ — contradicting minimality. So all vertices of W are distinct, i.e. W is a path. □

Lemma C (minimum degree 2 forces a cycle) *If G is finite and $\delta(G) \ge 2$, then G contains a cycle.*

Proof. G is finite, so among all paths of G there is one of maximum length, say $P = v_0v_1 \dots v_k$ (note $k \geq 1$, since $\delta \geq 2 > 0$ means G has an edge). Every neighbour of v_0 lies on P: a neighbour $y \notin V(P)$ would make $y\,v_0v_1 \dots v_k$ a strictly longer path. Since $d(v_0) \geq 2$ and one neighbour of v_0 is v_1, the vertex v_0 has another neighbour v_i with $2 \leq i \leq k$. Then $v_0v_1 \dots v_iv_0$ is a cycle, of length $i + 1 \geq 3$. □

> **Lemma D (components are degree-closed)** *Let H be a component of G and $x \in V(H)$. Then $N_G(x) \subseteq V(H)$, and consequently $d_H(x) = d_G(x)$.*

Proof. If $y \in N_G(x)$ then the edge xy joins y to x, so y lies in the same component as x, namely H. Hence $N_H(x) = N_G(x)$ and the two degrees agree. This innocuous remark is what allows us to apply counting results component by component. □

PROBLEM 1 Knights on the 3×3 board

a cyclic-order invariant

> **PROBLEM**
> Place two white and two dark knights in the corners of a 3×3 chessboard so that the identically coloured knights are on diametrically opposite corners. The knights move in the usual way known from chess, and two knights cannot stand on the same square at the same time. Is it possible to move the knights so that they stand in the corners at the end, but the diametrically opposite ones are of different colours?

Although it is posed as a chess puzzle, this is a question about a graph, and we treat it as one throughout. The squares a knight can stand on are the vertices; two squares are joined by an edge when a single knight's move carries one to the other. The four knights are then *tokens* occupying four distinct vertices, a legal move slides one token along an edge to an unoccupied vertex, and the question asked is which token configurations are reachable from which. Everything therefore hinges on the structure of that one graph — and it will turn out to be a single 8-cycle, which settles the matter at once.

The answer is **no**. We prove it by exhibiting a quantity — the cyclic order in which the four knights appear around a certain 8-cycle — that no legal move can ever change, and which differs between the starting position and every admissible target position.

Step 1. Set-up and the knight-move graph

Number the squares of the board

$$\begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{matrix}$$

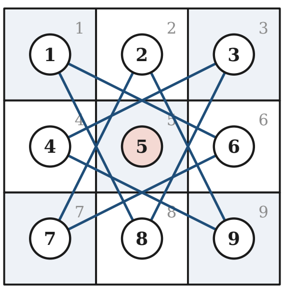
so that square s sits in row $\lfloor (s-1)/3 \rfloor$ and column $(s-1) \bmod 3$. The corners are $1, 3, 7, 9$; the two "diametrically opposite" corner pairs (opposite through the centre square 5) are $\{1, 9\}$ and $\{3, 7\}$.

Up to swapping the names of the two colours, the starting position is therefore

White on 1 and 9, Dark on 3 and 7.

The required final position is: all four knights on corners, with $\text{colour}(1) \neq \text{colour}(9)$ *and* $\text{colour}(3) \neq \text{colour}(7)$.

Define the *knight-move graph K*: its vertices are the nine squares, and two squares are adjacent when they are one knight's move apart, i.e. when their (row, column) vectors differ by $(\pm 1, \pm 2)$ or $(\pm 2, \pm 1)$. A knight standing on a square may move exactly along an edge of K, provided the target square is unoccupied. The complete list of moves is short enough to write down in full.



The eight knight moves available on the 3×3 board. The centre square 5 (shaded) is not the endpoint of any move.

Step 2. The usable squares form a single 8-cycle

Reading off the table of moves:

square	(row, col)	squares a knight can move to	degree
1	(0, 0)	6, 8	2
2	(0, 1)	7, 9	2
3	(0, 2)	4, 8	2
4	(1, 0)	3, 9	2
5	(1, 1)	— (none)	0
6	(1, 2)	1, 7	2
7	(2, 0)	2, 6	2
8	(2, 1)	1, 3	2
9	(2, 2)	2, 4	2

Table 1.1 — all knight moves on the 3×3 board. The centre square 5 is isolated; every other square has degree exactly 2.

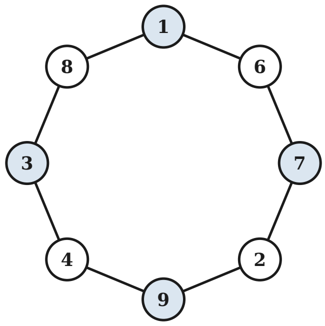
Two things are visible at once.

- The **centre square** 5 **is isolated** in K: from $(1, 1)$ every knight's move $(\pm 1, \pm 2)$ or $(\pm 2, \pm 1)$ leaves the 3×3 board. Hence no knight can ever move to 5 or away from 5. Since the knights start on corners, square 5 is empty at the start and stays empty forever; we may delete it.
- Every one of the remaining eight squares has degree **exactly** 2. So $K - 5$ is a 2-regular graph on 8 vertices, and it is connected: starting at 1 and repeatedly walking to the not-yet-used neighbour gives

$$1 - 6 - 7 - 2 - 9 - 4 - 3 - 8 - 1,$$

  a closed walk through all eight squares. Therefore $K - 5 \cong C_8$.

(This is the classical "straightening" of the puzzle: the tangle of knight moves on the board is really just a circle. It is also an instance of Problem 3 below: a connected 2-regular graph must be a cycle.)



The eight usable squares, redrawn as the 8-cycle 1 6 7 2 9 4 3 8. The four corners (shaded) sit in alternating positions around the cycle.

The crucial structural fact, plainly visible in the figure, is that the corners $1, 7, 9, 3$ occupy the four *alternate* vertices w_0, w_2, w_4, w_6 of the cycle $w_0 w_1 \dots w_7 = 1\,6\,7\,2\,9\,4\,3\,8$. In particular, when all four knights stand on corners, reading the cycle in its natural direction reads the corners in the cyclic order 1, 7, 9, 3.

Step 3. The invariant: four beads on a closed string

> **Claim** *Assign to each position the cyclic word obtained by listing the colours of the occupied vertices in the cyclic order of* C_8 *(a cyclic word of length* 4 *in the letters* W, D*, with two of each). This colour pattern is unchanged by every legal move.*

Proof. Fix a position. Let the four knights, listed in the cyclic order in which they occur along C_8, be k_1, k_2, k_3, k_4, and for $j = 1, 2, 3, 4$ let $a_j \geq 1$ be the number of cycle-edges on the arc that leads from k_j forward to k_{j+1} (indices mod 4). The four arcs tile the cycle, so

$$a_1 + a_2 + a_3 + a_4 \;=\; 8, \qquad a_j \;\geq\; 1 \text{ for all } j.$$

A legal move takes one knight, say k_j, to one of its two neighbours on C_8, and that neighbour must be *unoccupied*.

- If k_j steps *forward* (towards k_{j+1}), the square it steps onto is unoccupied exactly when the arc in front of it has length $a_j \geq 2$. After the move a_j becomes $a_j - 1 \;\geq 1$ and a_{j-1} becomes $a_{j-1} + 1$.
- If k_j steps *backward*, symmetrically $a_{j-1} \geq 2$ is required, and afterwards $a_{j-1} \mapsto a_{j-1} - 1 \geq 1$, $a_j \mapsto a_j + 1$.

In both cases all four arc lengths remain ≥ 1. An arc length of 0 is precisely what it would mean for one knight to catch up with and overtake the next one; since this never happens, the cyclic sequence k_1, k_2, k_3, k_4 — and hence the cyclic sequence of their colours — is literally the same list after the move as before. □

Informally: the knights are four beads threaded on a closed loop of string. A bead may slide one step at a time, but it can never jump over another bead, so the cyclic order of the beads is frozen for all time.

Step 4. The two patterns disagree

By Step 2, whenever all four knights stand on corners the colour pattern is read off the corners in the cyclic order 1, 7, 9, 3.

Starting position. $\mathrm{colour}(1) = \mathrm{W}$, $\mathrm{colour}(7) = \mathrm{D}$, $\mathrm{colour}(9) = \mathrm{W}$, $\mathrm{colour}(3) = \mathrm{D}$, giving the cyclic word

$$\mathrm{W\,D\,W\,D} \text{ — \textit{the two whites are separated by a dark knight on both sides.}}$$

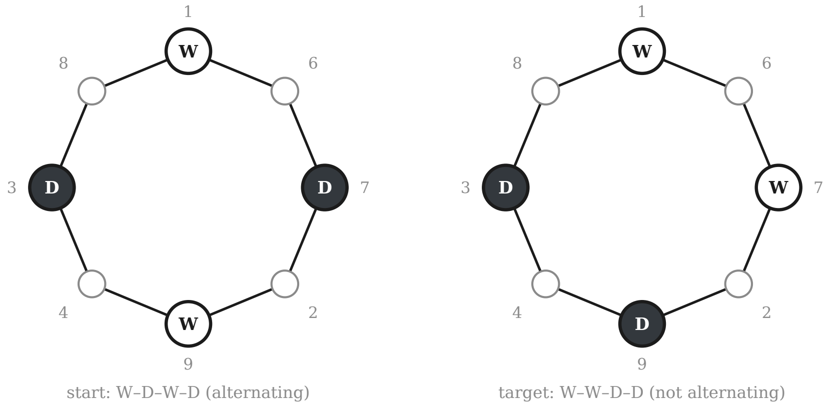
Any admissible target position. We need $\mathrm{colour}(1) \neq \mathrm{colour}(9)$ and $\mathrm{colour}(3) \neq \mathrm{colour}(7)$. As there are two knights of each colour this leaves exactly four possibilities, and all four give the same cyclic word:

colour(1)	colour(3)	colour(7)	colour(9)	read round the cycle: 1, 7, 9, 3	as a cyclic word
W	W	D	D	W D D W	WWDD
W	D	W	D	W W D D	WWDD
D	W	D	W	D D W W	WWDD
D	D	W	W	D W W D	WWDD

Table 1.2 — the four corner positions in which diametrically opposite knights have different colours. All four give the same cyclic word WWDD.

$$\text{In every case the target pattern is } \mathrm{W\,W\,D\,D} \text{ — \textit{the two whites are cyclically adjacent.}}$$

The cyclic words WDWD and WWDD are different: in the first, each white knight has a dark knight on either side; in the second, the two whites are neighbours in the cyclic order. No rotation of WDWD produces WWDD. By the Claim of Step 3 the pattern never changes, so no sequence of legal moves can lead from the starting position to any admissible target position.



Left: the start, pattern $\mathrm{W\,D\,W\,D}$. *Right: a required final position, pattern* $\mathrm{W\,W\,D\,D}$. *Small circles are empty squares. The two patterns are different cyclic words, and the pattern is invariant — hence the right-hand position is unreachable.*

ANSWER

No, it is not possible. The eight squares a knight can ever use form a single 8-cycle, on which the four knights behave like beads on a closed string: they can slide but never pass one another, so their cyclic order — and hence the cyclic pattern of their colours — is invariant. Starting from "identically coloured knights diametrically opposite" the pattern is WDWD, whereas every corner position with "diametrically opposite knights of different colours" has pattern WWDD.

Step 5. Exhaustive verification

A breadth-first search over all positions confirms the argument and shows that the invariant is *complete*, i.e. it is the only obstruction.

quantity	value
placements of 4 knights on the 8 usable squares (2 white, 2 dark)	420
... of these, the ones whose colour pattern is W D W D	**140**
positions actually reachable from the start by legal moves	**140** — exactly the ones above
reachable positions with all four knights on corners	2
... of these, positions with opposite corners of *different* colour	**0**
shortest solution of Guarini's puzzle (see remark)	16 moves

Table 1.3 — exhaustive breadth-first search over all positions. The colour pattern is not merely an obstruction, it is a complete invariant: every position with the right pattern is reachable.

The two reachable corner positions are the initial one and the one obtained from it by sliding all four knights two steps around the cycle (a 90° rotation of the board); in both, identically coloured knights are diametrically opposite, exactly as predicted. Moreover the number of reachable positions, 140, equals precisely the number of placements whose colour pattern is WDWD — every position with the correct pattern is reachable, and no other.

Remark (what *is* possible) Only the cyclic *order* is frozen; rotations are free. This is why the classical *Guarini puzzle* (1512) — white knights on the two top corners $1, 3$, dark knights on the two bottom corners $7, 9$, the task being to interchange them — *is* solvable: reading the corners in the cyclic order $1, 7, 9, 3$, the start is W D D W and the target is D W W D, and as cyclic words both equal WWDD. A shortest solution takes 16 moves. Guarini's puzzle and the present problem differ only in the initial arrangement, and that single difference decides solvability.

PROBLEM 2 Two degree sequences

Havel–Hakimi, Erdős–Gallai, parity

PROBLEM

Are there simple graphs with degree sequence **(a)** $1, 2, 2, 3, 3, 3,$ respectively **(b)** $1, 1, 2, 2, 3, 4, 4$?

A finite sequence of non-negative integers is called *graphical* if it is the degree sequence of some simple graph. Two necessary conditions are immediate for a sequence $d_1, \ldots, d_n$:

- **(N1) Range.** $0 \le d_i \le n-1$ for every i, since in a simple graph a vertex has at most $n-1$ distinct neighbours.
- **(N2) Parity.** $d_1 + \cdots + d_n$ is even, since by Lemma A it equals $2m$.

These are not sufficient in general (for instance $3, 3, 1, 1$ satisfies both but is not graphical), so for part (a) we shall need a genuine criterion. We use the Havel–Hakimi reduction, which has the advantage of *building* the graph, and then double-check with the Erdős–Gallai inequalities.

Part (a): $1,\ 2,\ 2,\ 3,\ 3,\ 3$

The two necessary conditions

Here $n = 6$ and the largest entry is $3 \le n - 1 = 5$, so (N1) holds. The sum is

$$1+2+2+3+3+3 \;=\; 14,$$

which is even, so (N2) holds and any such graph would have $m = \frac{14}{2} = 7$ edges. Both tests are passed; we now show the sequence really is graphical by constructing a graph.

The Havel–Hakimi reduction

> **Theorem (Havel 1955, Hakimi 1962)** *Let $d_1 \ge d_2 \ge \cdots \ge d_n \ge 0$ with $d_1 \le n-1$. Then $d_1, d_2, \ldots, d_n$ is graphical if and only if the sequence obtained by deleting d_1 and subtracting 1 from each of the next d_1 entries,*
>
> $$d_2 - 1,\ d_3 - 1,\ \ldots,\ d_{d_1+1} - 1,\ d_{d_1+2},\ \ldots,\ d_n,$$
>
> *(re-sorted into non-increasing order) is graphical.*

The "if" direction — the only one we need — is constructive and obvious: given a graph realising the reduced sequence, add a new vertex and join it to the d_1 vertices whose degrees were decreased. Applying the reduction repeatedly to our sequence, sorted as $3, 3, 3, 2, 2, 1$:

round	current sequence	operation	reduced sequence
1	`(3, 3, 3, 2, 2, 1)`	delete the leading 3; subtract 1 from the next 3 entries	`(2, 2, 2, 1, 1)`
2	`(2, 2, 2, 1, 1)`	delete the leading 2; subtract 1 from the next 2 entries	`(1, 1, 1, 1)`
3	`(1, 1, 1, 1)`	delete the leading 1; subtract 1 from the next 1 entry	`(1, 1, 0)`
4	`(1, 1, 0)`	delete the leading 1; subtract 1 from the next 1 entry	`(0, 0)`
5	`(0, 0)`	all entries are 0 — realised by the edgeless graph	**graphical ✓**

Table 2.1 — the Havel–Hakimi reduction of 3, 3, 3, 2, 2, 1 terminates in the zero sequence, so the sequence is graphical.

The reduction terminates in the all-zero sequence, which is realised by the edgeless graph. Hence $1, 2, 2, 3, 3, 3$ **is graphical**.

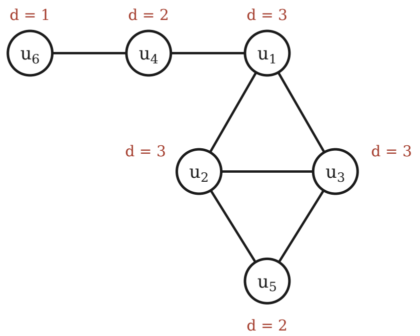
The graph the reduction produces

Reading the reduction backwards and naming the vertices $u_1, \ldots, u_6$ with target degrees $3, 3, 3, 2, 2, 1$ respectively, each round joins the current vertex of largest residual demand to the vertices of next largest residual demand:

- *Round 1:* u_1 (demand 3) is joined to u_2, u_3, u_4. Residual demands become $(u_2, u_3, u_4, u_5, u_6) = (2, 2, 1, 2, 1)$.
- *Round 2:* u_2 (demand 2) is joined to u_3 and u_5. Residual demands become $(u_3, u_4, u_5, u_6) = (1, 1, 1, 1)$.
- *Round 3:* the four remaining demands are all 1; pair them off as u_3u_5 and u_4u_6.

This yields the graph G^* with $V = \{u_1, \ldots, u_6\}$ and the seven edges

$$E(G^*) = \{\, u_1u_2,\ u_1u_3,\ u_1u_4,\ u_2u_3,\ u_2u_5,\ u_3u_5,\ u_4u_6 \,\}.$$



A simple graph with degree sequence $1, 2, 2, 3, 3, 3$: the "diamond" K_4 minus the edge u_1u_5 on $\{u_1, u_2, u_3, u_5\}$, with a pendant path $u_1 - u_4 - u_6$ attached.

Verification, vertex by vertex

G^* is simple (the seven listed pairs are distinct and none is a loop) and has $7 = \frac{14}{2}$ edges, as required. The degrees are:

vertex	its neighbours in G^*	degree
u_1	u_2, u_3, u_4	3
u_2	u_1, u_3, u_5	3
u_3	u_1, u_2, u_5	3
u_4	u_1, u_6	2
u_5	u_2, u_3	2
u_6	u_4	1
total		**14 = 2 · 7 ✓**

Table 2.2 — degrees in the constructed graph. Sorted, they read 1, 2, 2, 3, 3, 3, and the degree sum 14 = 2·7 confirms the edge count.

Sorted, the degrees read $1, 2, 2, 3, 3, 3$ — exactly the prescribed sequence.

Independent check: the Erdős–Gallai inequalities

> **Theorem (Erdős–Gallai 1960)** *A non-increasing sequence $d_1 \geq \cdots \geq d_n \geq 0$ with even sum is graphical if and only if for every $k \in \{1, \ldots, n\}$*
>
> $$\sum_{i=1}^{k} d_i \ \leq\ k(k-1)\ +\ \sum_{i=k+1}^{n} \min(d_i,\ k).$$

With $d = (3, 3, 3, 2, 2, 1)$ all six inequalities hold, comfortably:

k	$\Sigma_{i\le k}\, d_i$	$k(k-1) + \Sigma_{i>k} \min(d_i, k)$	holds?
1	3	0 + (1 + 1 + 1 + 1 + 1) = 5	✓
2	6	2 + (2 + 2 + 2 + 1) = 9	✓
3	9	6 + (2 + 2 + 1) = 11	✓
4	11	12 + (2 + 1) = 15	✓
5	13	20 + (1) = 21	✓
6	14	30 + (0) = 30	✓

Table 2.3 — the six Erdős–Gallai inequalities for 3, 3, 3, 2, 2, 1. All hold, independently confirming that the sequence is graphical.

Both criteria agree, and we have an explicit graph.

ANSWER (A)

Yes. The sequence $1, 2, 2, 3, 3, 3$ is graphical. One realisation is $E = \{u_1u_2, u_1u_3, u_1u_4, u_2u_3, u_2u_5, u_3u_5, u_4u_6\}$ on six vertices — a diamond (K_4 minus one edge) with a pendant path of length 2 attached, drawn above.

Part (b): 1, 1, 2, 2, 3, 4, 4

Here $n = 7$ and the largest entry is $4 \le n - 1 = 6$, so condition (N1) is satisfied and nothing suspicious is visible at first glance. But look at the sum:

$$1 + 1 + 2 + 2 + 3 + 4 + 4 \;=\; 17,$$

which is **odd**. By the handshake lemma (Lemma A, proved in Problem 4), in every graph

$$\sum_{v \in V(G)} d(v) \;=\; 2\,|E(G)|,$$

and the right-hand side is an even integer. A graph with this degree sequence would therefore satisfy $17 = 2m$, i.e. $m = 8.5$, which is absurd since m is a non-negative integer. Hence **no such graph exists**.

Two remarks on how robust this obstruction is.

- The argument never uses simplicity. Multiple edges are counted with multiplicity in the degree, and a loop contributes 2 to the degree of its vertex, so $\sum_v d(v) = 2m$ holds for multigraphs as well. So there is no *multigraph* with this degree sequence either — the sequence is unrealisable in the widest sense.
- Equivalently, one may invoke the corollary proved in Problem 4: *the number of odd-degree vertices is even.* The odd entries of $1, 1, 2, 2, 3, 4, 4$ are $1, 1, 3$ — there are three of them, an odd number, which is impossible.

ANSWER (B)

No. The entries sum to 17, an odd number, while $\sum_v d(v) = 2|E|$ is always even. Equivalently, the sequence has three odd entries, whereas every graph has an even number of odd-degree vertices. No graph — simple or not — has degree sequence $1, 1, 2, 2, 3, 4, 4$.

Remark (parity is the *only* defect in (b)) The failure is genuinely arithmetic, not structural: repairing the parity by a single small change makes the sequence realisable. For example $1, 1, 2, 2, \mathbf{2}, 4, 4$ (sum 16) and $1, 1, 2, 2, \mathbf{3}, 4, \mathbf{5}$ (sum 18) are both graphical, as the Havel–Hakimi reduction confirms. This is worth noticing because it shows condition (N2) doing real work: it is not a formality that some other property would have caught.

PROBLEM 3 2-regular graphs, and graphs with $\Delta(G) \le 2$

classification

PROBLEM

Determine the 2-regular graphs. What are those graphs that satisfy $\Delta(G) \le 2$?

Both classes admit a complete and very clean description:

- G is 2-regular $\iff$ G is a disjoint union of cycles;
- $\Delta(G) \le 2$ $\iff$ G is a disjoint union of paths and cycles.

We prove both, in that order; the second builds on the first.

Part 1. The 2-regular graphs

Theorem 3.1 *A finite simple graph G is 2-regular if and only if every component of G is a cycle; equivalently, if and only if G is a vertex-disjoint union $G \cong C_{n_1} \dot\cup C_{n_2} \dot\cup \ldots \dot\cup C_{n_k}$ of cycles, with $k \ge 1$ and every $n_i \ge 3$.*

Proof of "⇐". In the cycle C_m every vertex has exactly the two neighbours adjacent to it along the cycle, so C_m is 2-regular. Degrees are not affected by taking disjoint unions (a vertex acquires no new neighbours), so a disjoint union of cycles is 2-regular. □

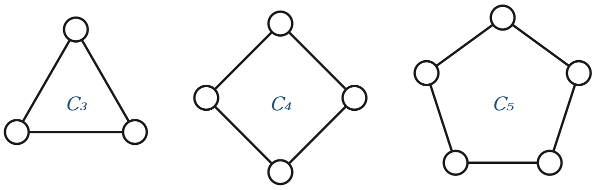
Proof of "⇒". Let G be 2-regular and let H be any component of G. By Lemma D, $d_H(x) = d_G(x) = 2$ for every $x \in V(H)$, so H is itself connected and 2-regular. In particular $\delta(H) = 2$, so Lemma C gives a cycle $C \subseteq H$.

We claim $H = C$. First, $V(H) = V(C)$. Suppose not; since H is connected and $V(C) \neq \emptyset$, there is an edge $xy \in E(H)$ with $x \in V(C)$ and $y \notin V(C)$. But x lies on the cycle C, so its two neighbours along C are already two distinct neighbours of x, both inside $V(C)$; as $d_H(x) = 2$ these are *all* of its neighbours, and $y \notin V(C)$ cannot be one of them — a contradiction. Hence $V(H) = V(C)$.

Second, $E(H) = E(C)$. Each $x \in V(C)$ already has its two C-neighbours, and $d_H(x) = 2$, so x is incident with no edge of H outside $E(C)$. As this holds for every vertex, $E(H) = E(C)$.

Therefore $H = C$ is a cycle, and G is the disjoint union of its components, each a cycle. Finally, in a *simple* graph a cycle has length at least 3 (a length-2 "cycle" would be a repeated edge, a length-1 one a loop), so every $n_i \geq 3$.

□



A typical 2-regular graph. Every 2-regular graph looks like this and nothing else.

A second, purely arithmetic proof of "⇒" Let H be a component, $|V(H)| = h$. By Lemma A applied to H, $2h = \sum_{v \in V(H)} d_H(v) = 2|E(H)|$, so $|E(H)| = h$: the component has exactly as many edges as vertices. A connected graph on h vertices has at least $h - 1$ edges, with equality exactly for trees; having h edges it contains exactly one cycle. Since $\delta(H) = 2$ forbids any vertex of degree 1, no tree-like appendage can hang off that cycle, so H is the cycle itself.

Corollary (counting) A 2-regular graph on n vertices has exactly n edges, and by Theorem 3.1 the 2-regular graphs on n vertices correspond bijectively (up to isomorphism) to the partitions of n into parts of size ≥ 3. For $n = 3, \ldots, 9$ the counts are $1, 1, 1, 2, 2, 3, 4$; e.g. for $n = 8$ the three graphs are C_8, $C_5 \dot\cup C_3$ and $C_4 \dot\cup C_4$. In particular there is no 2-regular simple graph on 1 or 2 vertices.

Part 2. The graphs with $\Delta(G) \leq 2$

> **Theorem 3.2** *A finite simple graph G satisfies $\Delta(G) \leq 2$ if and only if every component of G is a path (possibly the trivial path K_1) or a cycle. Equivalently: G is a vertex-disjoint union of paths and cycles.*

Proof of "⇐". In P_k the two end vertices have degree 1 and all others degree 2 (in $P_1 = K_1$ the only vertex has degree 0); in C_m every degree is 2. All these are ≤ 2, and disjoint unions do not change degrees. □

Proof of "⇒". Let H be a component of G; by Lemma D every vertex of H has $d_H = d_G \leq 2$. We distinguish two cases.

Case 1: every vertex of H has degree exactly 2. Then H is a connected 2-regular graph, so by Theorem 3.1 it is a cycle.

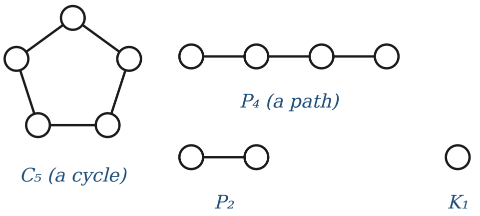
Case 2: some vertex $z \in V(H)$ has $d_H(z) \leq 1$. We show H is a path. If $|V(H)| = 1$ then $H = K_1$, the trivial path, and we are done; so assume $|V(H)| \geq 2$, whence $d_H(z) = 1$ (a vertex of degree 0 would be a component by itself). Build a walk greedily:

$$z = x_0, \quad x_1 := \text{the unique neighbour of } x_0, \quad x_{i+1} := \text{the neighbour of } x_i \text{ other than } x_{i-1} \ \ (i \geq 1),$$

continuing as long as x_i has a second neighbour, i.e. as long as $d_H(x_i) = 2$. The walk never revisits a vertex: if $x_j = x_i$ were the first repetition with $i < j$, then x_i would be incident with the three *distinct* edges $x_{i-1}x_i$, $x_i x_{i+1}$, $x_{j-1}x_j$ — distinct because $x_{j-1} \neq x_{i-1}$ and $x_{j-1} \neq x_{i+1}$ by the minimality of j — contradicting $d_H(x_i) \leq 2$. Hence the walk visits distinct vertices, and since H is finite it must stop, at a vertex x_k with $d_H(x_k) = 1$. So $P := x_0 x_1 \ldots x_k$ is a path in H.

Finally $H = P$. Every vertex of P has all of its H-neighbours on P: the interior vertices $x_1, \ldots, x_{k-1}$ have degree 2 with both neighbours x_{i-1}, x_{i+1} on P, and the two ends x_0, x_k have degree 1 with their unique neighbour on P. Therefore no edge leaves $V(P)$; since H is connected this forces $V(H) = V(P)$, and then $E(H) = E(P)$ because every vertex of P has already used up its degree inside P. So $H = P$ is a path.

□



A typical graph with $\Delta(G) \le 2$: components are paths (including isolated vertices) and cycles. Here $\Delta(G) = 2$ and $\delta(G) = 0$.

The counting version of Case 2 There is a slicker way to see that a tree T with $\Delta(T) \le 2$ is a path. Being acyclic and connected, T has $|E(T)| = |V(T)| - 1$; if $|V(T)| \ge 2$ then no degree is 0, so every degree is 1 or 2. Writing ℓ for the number of degree-1 vertices, Lemma A gives

$$\ell + 2\big(|V(T)| - \ell\big) \;=\; \sum_v d(v) \;=\; 2|E(T)| \;=\; 2|V(T)| - 2,$$

i.e. $2|V(T)| - \ell = 2|V(T)| - 2$, so $\ell = 2$ exactly: a tree with maximum degree ≤ 2 has precisely two leaves and is a path. (Case 2 of the proof above first checks that H is acyclic: a cycle in H would already use up both available degrees at each of its vertices, making H a cycle by the argument of Theorem 3.1 and contradicting the existence of z with $d_H(z) \le 1$.)

ANSWER

2-regular graphs are exactly the vertex-disjoint unions of cycles $C_{n_1} \dot\cup \cdots \dot\cup C_{n_k}$, $n_i \ge 3$; equivalently, graphs all of whose components are cycles. Such a graph has as many edges as vertices.

Graphs with $\Delta(G) \le 2$ are exactly the vertex-disjoint unions of paths and cycles — every component is a path (an isolated vertex being the trivial path) or a cycle. Comparing the two statements: G is 2-regular precisely when $\Delta(G) \le 2$ *and* $\delta(G) \ge 2$; relaxing $\Delta \le 2$ from regularity is exactly what admits the path components.

Remark (infinite graphs) If infinite graphs are allowed the classification acquires two extra items. A connected 2-regular graph may also be a *double ray* $\ldots v_{-2}v_{-1}v_0v_1v_2 \ldots$, the two-way infinite path; and a connected graph with $\Delta \le 2$ may in addition be a *ray* $v_0v_1v_2\ldots$ (one-way infinite), whose initial vertex has degree 1. So in general: components of a 2-regular graph are cycles or double rays, and components of a graph with $\Delta \le 2$ are finite paths, cycles, rays, or double rays. The ray reappears in Problem 4 as the counterexample to the odd-degree theorem.

PROBLEM 4 The number of odd-degree vertices is even

handshake lemma; the infinite case

PROBLEM

Show that the number of odd-degree vertices of a finite graph is even. Show that this is not necessarily true for infinite graphs.

Step 1. The handshake lemma

Theorem 4.1 (handshake lemma; Euler 1736) *For every finite graph $G = (V, E)$,*

$$\sum_{v \in V} d(v) \;=\; 2\,|E|.$$

Proof (double counting). Consider the set of *incidences*

$$I \;=\; \big\{\, (v, e) \in V \times E \;:\; v \text{ is an endpoint of } e \,\big\},$$

and count $|I|$, which is a finite number, in two different ways.

Grouping by edges. Every edge $e = xy \in E$ has exactly two endpoints, x and y, and they are distinct because G has no loops. So each edge contributes exactly 2 pairs to I, and

$$|I| \;=\; \sum_{e \in E} 2 \;=\; 2|E|.$$

Grouping by vertices. For a fixed vertex v, the pairs $(v, e) \in I$ correspond exactly to the edges incident with v, and there are $d(v)$ of them — this is the definition of the degree (in a simple graph the edges at v are in bijection with the neighbours of v). So

$$|I| \;=\; \sum_{v \in V} d(v).$$

The two counts evaluate the same finite number, which gives the identity.

□

The name comes from the standard reformulation: at a party, the total number of hands shaken, counted over all guests, is twice the number of handshakes — because each handshake involves exactly two hands. The lemma also holds verbatim for multigraphs (parallel edges are counted with multiplicity at both endpoints), and for graphs with loops under the usual convention that a loop at v contributes 2 to $d(v)$ — which is precisely the convention that keeps the double count correct, since a loop yields two incidences at its single vertex.

Step 2. The parity corollary

> **Corollary 4.2** *In every finite graph G the number of vertices of odd degree is even.*

Proof. Partition the vertex set according to the parity of the degree,

$$V \;=\; V_{\mathrm{odd}} \,\dot\cup\, V_{\mathrm{even}}, \qquad V_{\mathrm{odd}} = \{v : d(v) \text{ odd}\}, \quad V_{\mathrm{even}} = \{v : d(v) \text{ even}\},$$

and split the sum of Theorem 4.1 accordingly:

$$2|E| \;=\; \sum_{v\in V} d(v) \;=\; \underbrace{\sum_{v\in V_{\mathrm{odd}}} d(v)}_{=:S_{\mathrm{odd}}} \;+\; \underbrace{\sum_{v\in V_{\mathrm{even}}} d(v)}_{=:S_{\mathrm{even}}}.$$

Now $2|E|$ is even, and S_{even} is a finite sum of even numbers, hence even. Therefore

$$S_{\mathrm{odd}} \;=\; 2|E| - S_{\mathrm{even}}$$

is even as well. On the other hand S_{odd} is a sum of $|V_{\mathrm{odd}}|$ *odd* numbers, so reducing modulo 2, where every odd number is $\equiv 1$,

$$0 \;\equiv\; S_{\mathrm{odd}} \;= \sum_{v\in V_{\mathrm{odd}}} d(v) \;\equiv\; \sum_{v\in V_{\mathrm{odd}}} 1 \;=\; |V_{\mathrm{odd}}| \pmod{2}.$$

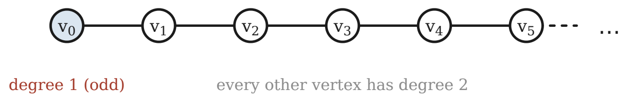
Hence $|V_{\mathrm{odd}}|$ is even. □

Three quick illustrations: K_4 has all four degrees equal to 3, and 4 is even; the path P_3 has degrees $1, 2, 1$, so two odd vertices; the sequence $1, 1, 2, 2, 3, 4, 4$ of Problem 2(b) has the three odd entries $1, 1, 3$, an odd number of them — which is exactly why no graph realises it.

Step 3. Infinite graphs: a counterexample

Corollary 4.2 fails for infinite graphs, and the smallest possible example already does the job.

The ray R Let $V(R) = \{v_0, v_1, v_2, \ldots\}$ be a countably infinite vertex set and $E(R) = \{\, v_i v_{i+1} : i \geq 0 \,\}$. This one-way infinite path is called a *ray*.



The ray R: a connected infinite graph with exactly one vertex of odd degree.

The degrees are immediate from the definition:

$$d(v_0) = 1 \quad (\text{only the neighbour } v_1), \qquad\qquad d(v_i) = 2 \quad (\text{the neighbours } v_{i-1}, v_{i+1}) \text{ for } i \geq 1.$$

Hence $V_{\mathrm{odd}}(R) = \{v_0\}$ and

$$|V_{\mathrm{odd}}(R)| \;=\; 1,$$

which is **odd**. So the number of odd-degree vertices of an infinite graph need not be even. R is even connected and locally finite (all degrees are finite), so no pathology beyond infiniteness itself is being exploited.

Where exactly does the proof break down?

In R both sides of the handshake identity are infinite: $|E(R)| = \aleph_0$ and $\sum_v d(v) = 1 + 2 + 2 + 2 + \cdots = \infty$. The identity is not *false* — as cardinal arithmetic it still reads $\aleph_0 = 2 \cdot \aleph_0$ — it has simply become *vacuous*. The proof of Corollary 4.2 used the finiteness of $2|E|$ in an essential way, at the step "$2|E|$ is an even integer, and $S_{\mathrm{odd}} = 2|E| - S_{\mathrm{even}}$": one cannot subtract infinite quantities, and an infinite cardinal is neither even nor odd. Every trace of parity information is lost.

Counterexamples with any odd number of odd vertices

The construction generalises immediately. For $k \geq 1$ let $R^{(k)}$ be the disjoint union of k copies of the ray R. Each copy contributes exactly one vertex of degree 1, so $R^{(k)}$ has exactly k vertices of odd degree. Choosing k odd gives infinitely many counterexamples; choosing $k = 1$ gives the connected one above.

ANSWER

Finite case. By double counting the vertex–edge incidences, $\sum_v d(v) = 2|E|$ is even. Splitting this sum over the odd-degree and even-degree vertices, the even-degree part is even, so the odd-degree part is even too; being a sum of $|V_{\text{odd}}|$ odd numbers, it is $\equiv |V_{\text{odd}}| \pmod 2$. Hence $|V_{\text{odd}}|$ is even.

Infinite case. False in general. The ray $v_0v_1v_2\dots$ has $d(v_0) = 1$ and $d(v_i) = 2$ for $i \geq 1$, so it has exactly *one* vertex of odd degree. The proof collapses because $\sum_v d(v) = 2|E| = \infty$ carries no parity information.

Remark (what survives) It is the infinitude of the *edge* set, not of the vertex set, that destroys the result. If G is an arbitrary graph — possibly with infinitely many vertices — but with only *finitely many edges*, then Corollary 4.2 still holds: only finitely many vertices are incident with an edge, all other vertices have degree 0 (even), and applying the corollary to the finite subgraph spanned by the edges settles the matter. The ray, by contrast, has infinitely many edges.

PROBLEM 5 An odd-degree vertex is joined to another one

consequence of Problem 4

PROBLEM

Prove that if the degree of a vertex $u \in V(G)$ is odd, then there exists a u–v path in G such that $u \neq v$ and the degree of v is also odd.

The point of the statement is that a vertex of odd degree cannot be "odd in isolation": somewhere in its own connected component there must be a second odd-degree vertex, and being in the same component it is reachable by a path. The proof is Corollary 4.2 applied not to G but to a single component of G — and Lemma D is what makes that legitimate.

Theorem 5.1 *Let G be a finite graph and let $u \in V(G)$ have odd degree. Then there is a vertex $v \in V(G)$ with $v \neq u$ and $d_G(v)$ odd, and a u–v path in G.*

Proof. Let H be the connected component of G that contains u. Since G is finite, so is H, and H is a graph in its own right, so Corollary 4.2 applies to it:

$$\left|\{\, x \in V(H) \;:\; d_H(x) \text{ is odd} \,\}\right| \text{ is even.} \tag{5.1}$$

Degrees in H are degrees in G. By Lemma D, every neighbour in G of a vertex $x \in V(H)$ again lies in H; hence $d_H(x) = d_G(x)$ for all $x \in V(H)$. So the set counted in (5.1) is exactly

$$A \;:=\; \{\, x \in V(H) : d_G(x) \text{ is odd} \,\}, \qquad \text{and } |A| \text{ is even.}$$

A is non-empty, hence has at least two elements. Indeed $u \in V(H)$ and $d_G(u)$ is odd by hypothesis, so $u \in A$ and $|A| \geq 1$. But $|A|$ is even while 1 is odd, so $|A| \neq 1$; hence $|A| \geq 2$. Choose any $v \in A \setminus \{u\}$. Then $v \neq u$ and $d_G(v)$ is odd.

There is a u–v path. Both u and v lie in $V(H)$, and H is connected, so there is a u–v walk inside H. By Lemma B that walk contains a u–v path, which is a path of $H \subseteq G$, hence a path of G.

This produces the required vertex v and the required path.

□

ANSWER

Apply Corollary 4.2 to the *component* H containing u, not to G as a whole. Since a component contains all neighbours of each of its vertices, degrees inside H coincide with degrees in G, so H has an even number of odd-degree vertices. That number is at least 1 because u is one of them; being even it is at least 2, so H contains a second odd-degree vertex $v \neq u$. Since H is connected there is a u–v walk in H, and every walk contains a path (Lemma B). This gives the required u–v path with both endpoints of odd degree.

Comments on the hypotheses

Why "$u \neq v$" is the whole content Without the requirement $v \neq u$ the statement would be trivial: take $v = u$ and the path of length 0. The substance of the theorem is that a *second*, different odd vertex must exist, and that it can be reached without leaving the component.

Why passing to the component is necessary It would not be enough to apply Corollary 4.2 to G itself. That gives another odd-degree vertex $v \neq u$ somewhere in G, but there is no reason for it to be joined to u by a path: consider $G = P_2 \mathbin{\dot\cup} P_2$, two disjoint edges. All four vertices have odd degree 1, but the two odd vertices in the "other" copy are unreachable from u. Localising the counting argument to the component of u is exactly what rules this out.

Finiteness is essential As in Problem 4, the statement fails for infinite graphs, and the ray $R = v_0v_1v_2\dots$ is again the counterexample. R is connected, $u = v_0$ has odd degree 1, and v_0 is joined by a path to every other vertex — yet *every* other vertex has degree 2, which is even. So no v as in the theorem exists.

A classical strengthening The proof shows a little more than it claims: in *each* component the odd-degree vertices occur in even number, so the odd-degree vertices of a finite graph can be split into pairs, each pair joined by a path inside its own component. A well-known theorem of Euler–Listing type sharpens this considerably: a connected graph with exactly $2k > 0$ vertices of odd degree can be decomposed into k pairwise *edge-disjoint* trails, each of which joins two odd-degree vertices. For $k = 0$ this is the Euler-circuit theorem, and for $k = 1$ the Euler-trail theorem. We do not need — and do not prove — this sharper statement here; it is mentioned only to place Theorem 5.1 in context.

PROBLEM 6 Which of the three drawings are isomorphic?

circulant graphs; a counting invariant

PROBLEM

Determine which of the three graphs shown in the figure are isomorphic to each other.

Step 1. Reading the three drawings

Each of the three pictures shows 9 vertices drawn at the corners of a regular nonagon. Label the positions $0, 1, \ldots, 8$ clockwise, starting at the top vertex, and identify the label set with the cyclic group $\mathbb{Z}_9 = \{0, 1, \ldots, 8\}$ of residues modulo 9. For two positions i, j call

$$\operatorname{dist}(i, j) \;=\; \min\big((j - i) \bmod 9,\; (i - j) \bmod 9\big) \;\in\; \{1, 2, 3, 4\}$$

their *circular distance*; it is the number of steps between them around the nonagon, and it is exactly what determines how a chord looks in the drawing. Reading off each picture, every drawn edge joins two vertices at a fixed pair of circular distances, and each of the three graphs turns out to be a **circulant graph**: writing

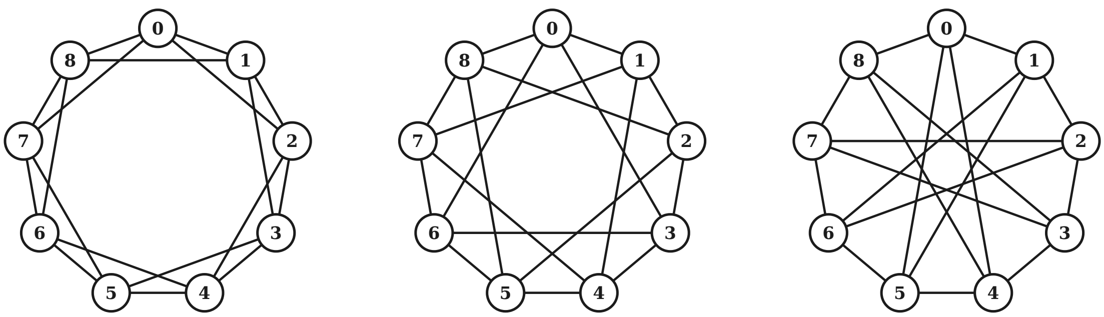
$$C_9(s, t) \;=\; \big(\,\mathbb{Z}_9,\; \{\, ij : \operatorname{dist}(i, j) \in \{s, t\}\,\}\,\big),$$

the three drawings are

drawing	name	what the picture shows	circular distances	graph
first (left)	G_1	the nonagon + the 9 chords skipping one vertex	$\{1, 2\}$	$C_9(1, 2)$
second (middle)	G_2	the nonagon + the 9 chords skipping two vertices (three triangles)	$\{1, 3\}$	$C_9(1, 3)$
third (right)	G_3	the nonagon + the 9 long chords skipping three vertices	$\{1, 4\}$	$C_9(1, 4)$

Table 6.1 — identifying the three drawings as circulant graphs on $\mathbb{Z}_9$.

Indeed the first drawing consists of the nonagon itself (the 9 edges of distance 1) together with the 9 "short" chords skipping one vertex (distance 2); the second has the nonagon plus the 9 chords of distance 3, which form three large superimposed triangles; the third has the nonagon plus the 9 long chords of distance 4, which pass close to the centre and produce the star pattern.



$G_1 = C_9(1,2)$ $G_2 = C_9(1,3)$ $G_3 = C_9(1,4)$

The three graphs redrawn with vertices labelled $0, \ldots, 8$ clockwise from the top. Each is 4-regular on 9 vertices with 18 edges.

The complete edge lists, sorted by circular distance, are:

graph	the 9 edges of distance 1 (the nonagon)	the 9 chords	$\lvert E \rvert$
G_1	`01 08 12 23 34 45 56 67 78`	`02 07 13 18 24 35 46 57 68`	18
G_2	`01 08 12 23 34 45 56 67 78`	`03 06 14 17 25 28 36 47 58`	18
G_3	`01 08 12 23 34 45 56 67 78`	`04 05 15 16 26 27 37 38 48`	18

Table 6.2 — complete edge lists; the pair `ij` denotes the edge $\{i, j\}$. Each graph has 18 edges, so each is 4-regular.

Step 2. The cheap invariants do not separate them

In each $C_9(s,t)$ with $s \neq t$ and $s,t \in \{1,2,3,4\}$, every vertex i has exactly the four neighbours $i \pm s,\ i \pm t$, which are pairwise distinct modulo 9. Hence all three graphs are 4-regular with

$$|V| = 9, \qquad |E| \ = \ \tfrac{1}{2}\sum_v d(v) \ = \ \tfrac{9\cdot 4}{2} \ = \ 18.$$

All three are connected (each contains the spanning nonagon of distance-1 edges) and each contains a triangle, so girth $= 3$ throughout. Every one of the usual first-line invariants therefore agrees, and we must look harder.

graph	|V|	|E|	degrees	connected	girth	triangles	4-cycles
$G_1 = C_9(1,2)$	9	18	4-regular	yes	3	**9**	**9**
$G_2 = C_9(1,3)$	9	18	4-regular	yes	3	**3**	**18**
$G_3 = C_9(1,4)$	9	18	4-regular	yes	3	**9**	**9**

Table 6.3 — the first six invariants agree for all three graphs; the triangle count (and the 4-cycle count) is what finally separates G_2 from the other two.

Step 3. $G_1 \cong G_3$: an explicit isomorphism

Theorem 6.1 *The map $\varphi : \mathbb{Z}_9 \to \mathbb{Z}_9$, $\varphi(i) = 4i \bmod 9$, is an isomorphism $G_1 = C_9(1,2) \ \longrightarrow \ G_3 = C_9(1,4)$.*

Proof.
(i) φ is a bijection. Since $\gcd(4,9) = 1$, the residue 4 is invertible modulo 9; concretely $4 \cdot 7 = 28 = 3 \cdot 9 + 1 \equiv 1 \pmod 9$, so $\varphi^{-1}(j) = 7j \bmod 9$. Explicitly:

i	0	1	2	3	4	5	6	7	8
$\varphi(i) = 4i \bmod 9$	**0**	**4**	**8**	**3**	**7**	**2**	**6**	**1**	**5**
$\varphi^{-1}(i) = 7i \bmod 9$	0	7	5	3	1	8	6	4	2

Table 6.4 — the isomorphism φ and its inverse, as permutations of $\mathbb{Z}_9$. Note φ fixes 0, 3 and 6.

(ii) φ maps edges to edges. Let $ij \in E(G_1)$, i.e. $j - i \equiv \pm 1$ or $\pm 2 \pmod 9$. Multiplying by 4,

$$\varphi(j) - \varphi(i) \ = \ 4(j-i) \ \equiv \ \begin{cases} \pm 4 \pmod 9, & \text{if } j - i \equiv \pm 1, \\ \pm 8 \equiv \mp 1 \pmod 9, & \text{if } j - i \equiv \pm 2, \end{cases}$$

using $8 \equiv -1$. In both cases the difference is ± 1 or ± 4, i.e. the circular distance of $\varphi(i)$ and $\varphi(j)$ lies in $\{1,4\}$, so $\varphi(i)\varphi(j) \in E(G_3)$. Note also $\varphi(i) \neq \varphi(j)$ by (i), so no edge degenerates.

*(iii) φ maps edges onto **all** edges.* By (i) and (ii) the induced map $E(G_1) \to E(G_3)$, $\{i,j\} \mapsto \{\varphi(i), \varphi(j)\}$, is injective; since $|E(G_1)| = |E(G_3)| = 18$ is finite, it is a bijection. (Alternatively, argue directly with φ^{-1}: $7 \cdot (\pm 1) = \pm 7 \equiv \mp 2$ and $7 \cdot (\pm 4) = \pm 28 \equiv \pm 1$, so φ^{-1} carries distances $\{1,4\}$ back to distances $\{2,1\}$.)

A bijection of vertex sets that maps the edge set onto the edge set is an isomorphism.

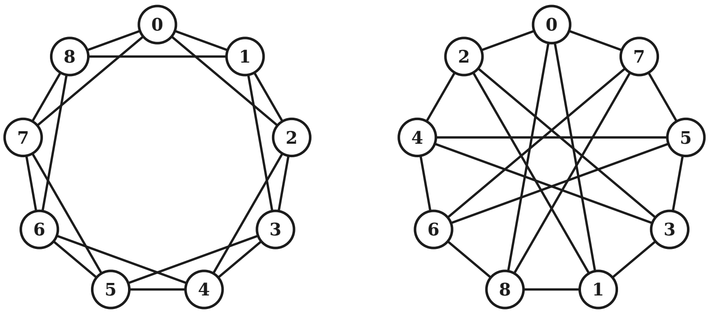
$\square$

Because the problem asks for every step, here is the induced map on all 18 edges of G_1, each verified to land on an edge of G_3:

edge of G_1	d		image	d	in G_3?	edge of G_1	d		image	d	in G_3?
{0,1}	1	→	{0,4}	4	✓	{3,4}	1	→	{3,7}	4	✓
{0,2}	2	→	{0,8}	1	✓	{3,5}	2	→	{2,3}	1	✓
{0,7}	2	→	{0,1}	1	✓	{4,5}	1	→	{2,7}	4	✓
{0,8}	1	→	{0,5}	4	✓	{4,6}	2	→	{6,7}	1	✓
{1,2}	1	→	{4,8}	4	✓	{5,6}	1	→	{2,6}	4	✓
{1,3}	2	→	{3,4}	1	✓	{5,7}	2	→	{1,2}	1	✓
{1,8}	2	→	{4,5}	1	✓	{6,7}	1	→	{1,6}	4	✓
{2,3}	1	→	{3,8}	4	✓	{6,8}	2	→	{5,6}	1	✓
{2,4}	2	→	{7,8}	1	✓	{7,8}	1	→	{1,5}	4	✓

Table 6.6 — every one of the 18 edges of G_1, and its image under φ. Distance 1 becomes distance 4, distance 2 becomes distance 8 ≡ −1: in all cases an edge of $G_3 = C_9(1,4)$.

All 18 images are distinct edges of G_3, and G_3 has exactly 18 edges — so the edge sets correspond perfectly.



The isomorphism made visible. Left: $G_1 = C_9(1,2)$ drawn with vertex i in position i. Right: the same graph G_1, with vertex i drawn in position $\varphi(i) = 4i$. The right-hand picture is the drawing of $G_3 = C_9(1,4)$ — only the labels have travelled.

Step 4. G_2 is isomorphic to neither: counting triangles

The number of triangles (subgraphs isomorphic to K_3) is an isomorphism invariant: an isomorphism $f : G \to G'$ maps any triangle $\{x, y, z\}$ of G to a triangle $\{f(x), f(y), f(z)\}$ of G', and f^{-1} maps triangles back, so the two collections of triangles are in bijection. We count the triangles of each $C_9(s,t)$ by hand.

The method. Let $S \subseteq \mathbb{Z}_9 \setminus \{0\}$ be the *connection set*, i.e. $i \sim j \iff j - i \in S$; note $S = -S$. For $C_9(1,2)$, $C_9(1,3)$, $C_9(1,4)$ these are, as residues,

$$S_1 = \{1,2,7,8\}, \qquad S_2 = \{1,3,6,8\}, \qquad S_3 = \{1,4,5,8\}$$

(recall $-1 \equiv 8,\ -2 \equiv 7,\ -3 \equiv 6,\ -4 \equiv 5$). A triangle is a set $\{x, y, z\}$; writing $a = y - x,\ b = z - y,\ c = x - z$ we must have

$$a, b, c \in S \qquad \text{and} \qquad a + b + c \equiv 0 \pmod 9.$$

Since $a, b, c \in \{1, \ldots, 8\}$, the integer $a + b + c$ lies between 3 and 24, so the congruence means $a + b + c \in \{9, 18\}$. So we must list the multisets $\{a, b, c\}$ drawn from S whose sum is 9 or 18.

Sum 18 comes free. Because $S = -S$, the map $x \mapsto 9 - x$ is a bijection of S onto itself, and it turns a solution $\{a, b, c\}$ with $a + b + c = 9$ into the solution $\{9 - a, 9 - b, 9 - c\}$ with sum $27 - 9 = 18$, and back again. So the sum-18 solutions are exactly the negatives of the sum-9 solutions, and they describe the same triangles traversed the other way round. It is therefore enough to solve $a + b + c = 9$.

Solving $a + b + c = 9$. Fix one element $a \in S$; the other two must be elements of S summing to $9 - a$. So we first list, once and for all, which numbers arise as a sum of two elements of S (repetitions allowed):

graph	connection set S	all sums $b + c$ with $b, c \in S$	required partner sums $9 - a$, $a \in S$
G_1	{1, 2, 7, 8}	2, 3, 4, 8, 9, 10, 14, 15, 16	8, 7, 2, 1
G_2	{1, 3, 6, 8}	2, 4, 6, 7, 9, 11, 12, 14, 16	8, 6, 3, 1
G_3	{1, 4, 5, 8}	2, 5, 6, 8, 9, 10, 12, 13, 16	8, 5, 4, 1

Table 6.5 — a triangle needs a + b + c ≡ 0 (mod 9) with a, b, c ∈ S. Comparing the third column with the fourth decides every case at a glance.

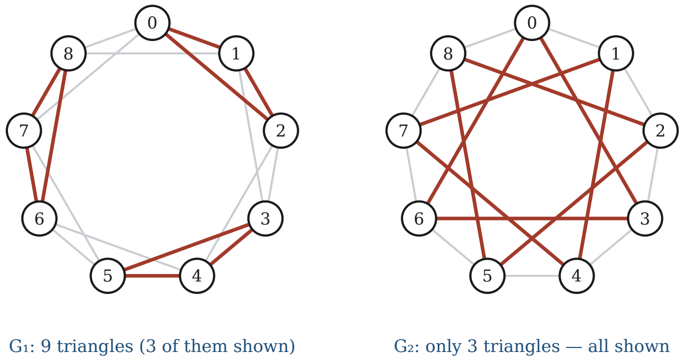
Now run through $a \in S$ in each case.

- **G_1, $S_1 = \{1,2,7,8\}$.** The required partner sums $9 - a$ are $8, 7, 2, 1$ for $a = 1, 2, 7, 8$. Comparing with the achievable list $\{2,3,4,8,9,10,14,15,16\}$: 8 is achievable (as $1 + 7$), 7 is not, 2 is achievable (as $1 + 1$), 1 is not. The two hits give the multisets $\{1,1,7\}$ and $\{7,1,1\}$ — the same multiset. So $\{1,1,7\}$ is the only solution. It realises the triangles $\{x,\ x+1,\ x+2\}$: the three differences are 1, 1 and $x - (x+2) = -2 \equiv 7$. As x runs through $\mathbb{Z}_9$ these nine sets of three consecutive residues are pairwise distinct. **9 triangles.**
- **G_2, $S_2 = \{1,3,6,8\}$.** The required partner sums are $8, 6, 3, 1$ for $a = 1, 3, 6, 8$. The achievable list is $\{2,4,6,7,9,11,12,14,16\}$: 8 no, 6 yes (as $3 + 3$), 3 no, 1 no. The single hit gives $\{3,3,3\}$, realising the triangles $\{x,\ x+3,\ x+6\}$. But here the nine values of x produce only *three* distinct sets, namely $\{0,3,6\}, \{1,4,7\}, \{2,5,8\}$ — the cosets of the subgroup $\{0,3,6\} \leq \mathbb{Z}_9$, each of which is counted three times as x runs over it. **3 triangles.**
- **G_3, $S_3 = \{1,4,5,8\}$.** The required partner sums are $8, 5, 4, 1$ for $a = 1, 4, 5, 8$. The achievable list is $\{2,5,6,8,9,10,12,13,16\}$: 8 yes (as $4 + 4$), 5 yes (as $1 + 4$), 4 no, 1 no. Both hits give the same multiset $\{1,4,4\}$, realising the triangles $\{x,\ x+1,\ x+5\}$: the differences are 1, 4 and $x - (x+5) = -5 \equiv 4$. These nine sets are pairwise distinct (they are not unions of cosets of any subgroup: $\{0,1,5\}, \{1,2,6\}, \ldots$ all differ). **9 triangles.**

The decisive difference is visible in the middle case: in G_2 the connection distance 3 is a *zero-divisor* modulo 9 — the element 3 generates only the three-element subgroup $\{0,3,6\}$ instead of all of $\mathbb{Z}_9$ — so the nine "triangle translates" collapse onto just three sets. In G_1 and G_3 no such collapse occurs.

graph	connection set S	multisets in S with sum 9	... with sum 18	the triangles	how many
$G_1 = C_9(1,2)$	{1, 2, 7, 8}	{1, 1, 7}	{2, 8, 8}	{x, x+1, x+2}	**9**
$G_2 = C_9(1,3)$	{1, 3, 6, 8}	{3, 3, 3}	{6, 6, 6}	{x, x+3, x+6}	**3**
$G_3 = C_9(1,4)$	{1, 4, 5, 8}	{1, 4, 4}	{5, 5, 8}	{x, x+1, x+5}	**9**

Table 6.7 — solving a + b + c ≡ 0 (mod 9) with a, b, c ∈ S. In G_2 the only solution is {3,3,3}, and {x, x+3, x+6} takes just three distinct values — the cosets of {0, 3, 6}. In the other two graphs the nine values are all distinct.



Left: G_1 contains the nine triangles $\{x, x+1, x+2\}$ (three are highlighted). Right: all of G_2's triangles — just the three cosets of $\{0, 3, 6\}$. Nine versus three: the graphs cannot be isomorphic.

Corollary 6.2 *$G_2 \ncong G_1$ and $G_2 \ncong G_3$.*

Proof. G_1 and G_3 contain 9 triangles each, while G_2 contains only 3. Since the number of triangles is preserved by isomorphism and $9 \neq 3$, neither G_1 nor G_3 can be isomorphic to G_2. □

ANSWER

The first and the third graphs are isomorphic; the second is isomorphic to neither. Labelling the nine vertices $0, \ldots, 8$ clockwise from the top, the drawings are the circulant graphs $G_1 = C_9(1,2)$, $G_2 = C_9(1,3)$ and $G_3 = C_9(1,4)$. An explicit isomorphism $G_1 \to G_3$ is multiplication by 4 modulo 9,

$$\varphi(i) = 4i \bmod 9 : \quad 0 \mapsto 0,\ 1 \mapsto 4,\ 2 \mapsto 8,\ 3 \mapsto 3,\ 4 \mapsto 7,\ 5 \mapsto 2,\ 6 \mapsto 6,\ 7 \mapsto 1,\ 8 \mapsto 5,$$

which turns distances 1 into 4 and distances 2 into $8 \equiv -1$. And G_2 is different from both because it contains only 3 triangles, whereas G_1 and G_3 contain 9.

Further remarks

The general principle behind φ For any n and any unit $u \in \mathbb{Z}_n^\times$, the multiplication map $x \mapsto ux$ is an isomorphism $C_n(S) \to C_n(uS)$ — it is an automorphism of the group $\mathbb{Z}_n$, so it turns differences into u-multiples of differences. Here $n = 9$, $u = 4$, and $4 \cdot \{1,2\} = \{4,8\}$, which as circular distances is $\{4,1\}$. The units modulo 9 are $\{1,2,4,5,7,8\}$, and applying each to $\{1,3\}$ yields only $\{1,3\}, \{2,3\}, \{3,4\}$: the distance 3 is never removed, because $\gcd(3,9) = 3 \neq 1$ means 3 generates the proper subgroup $\{0,3,6\}$ — which is precisely the source of G_2's three triangles. (Note that this observation alone only shows that no *multiplier* works; the triangle count in Step 4 is what proves that no isomorphism whatsoever exists.)

A second separating invariant, and a curiosity Triangles are not the only witness: counting 4-cycles gives 9, 18, 9 for G_1, G_2, G_3 respectively, separating G_2 just as well. Finally, a pleasant coincidence: the complement of G_1 is $C_9(3,4)$, and multiplying by the unit 2 sends $\{3,4\}$ to $\{6,8\} = \{-3,-1\}$, i.e. to distances $\{1,3\}$. So $\overline{G_1} \cong C_9(1,3) = G_2$: the odd one out is exactly the complement of the other two (which is consistent with 4-regularity, since $9 - 1 - 4 = 4$).

Exhaustive machine check All $9! = 362\,880$ bijections $V \to V$ were tested for each of the three pairs. The result confirms the conclusions above: $G_1 \cong G_3$ with exactly 18 distinct isomorphisms between them (so $|\mathrm{Aut}(G_1)| = 18$, the dihedral group of the nonagon, of order $2 \cdot 9$), and no isomorphism at all in the pairs (G_1, G_2) and (G_2, G_3). The map $\varphi(i) = 4i$ is among the 18.

PROBLEM 7 Minimum degree $n/2$ forces connectivity

two proofs; sharpness

PROBLEM

Let G be a simple graph and assume that $d(v) \geq \frac{n}{2}$ holds for all of its vertices. Prove that G is connected.

Here $n = |V(G)|$ is the number of vertices, so the hypothesis is $\delta(G) \geq n/2$. We give two proofs. The first is the shortest; the second costs one extra line and yields the stronger conclusion $\operatorname{diam}(G) \leq 2$ — any two vertices are joined by a path of length at most 2.

> **Theorem 7.1** *Let G be a simple graph on $n \geq 1$ vertices with $d(v) \geq n/2$ for every $v \in V(G)$. Then G is connected, and moreover $\operatorname{diam}(G) \leq 2$.*

Proof 1 — via the size of a smallest component

Proof. Suppose, for contradiction, that G is *not* connected. Then G has at least two components; let H_1 and H_2 be two distinct ones and let H be the smaller of the two, say with $h = |V(H)|$ vertices. Distinct components are vertex-disjoint subsets of $V(G)$, so

$$2h \;\leq\; |V(H_1)| + |V(H_2)| \;\leq\; n, \qquad \text{i.e.} \qquad h \;\leq\; \frac{n}{2}.$$

Pick any vertex $v \in V(H)$. By Lemma D all neighbours of v lie in H, and v is not its own neighbour (G is simple, so there are no loops); hence

$$d(v) \;=\; d_H(v) \;\leq\; |V(H)| - 1 \;=\; h - 1 \;\leq\; \frac{n}{2} - 1 \;<\; \frac{n}{2}.$$

This contradicts the hypothesis $d(v) \geq n/2$. Therefore G is connected. □

Proof 2 — via a common neighbour

Proof. Let $u \neq v$ be any two vertices of G; we exhibit a u–v path of length at most 2.

Case 1: $uv \in E(G)$. Then uv is itself a u–v path of length 1.

Case 2: $uv \notin E(G)$. We claim $N(u) \cap N(v) \neq \emptyset$. First locate the two neighbourhoods. Since G is simple there are no loops, so $u \notin N(u)$ and $v \notin N(v)$; and since $uv \notin E(G)$ we also have $v \notin N(u)$ and $u \notin N(v)$. Hence both $N(u)$ and $N(v)$ are contained in $V(G) \setminus \{u, v\}$, and therefore

$$\big|N(u) \cup N(v)\big| \;\leq\; n - 2.$$

By inclusion–exclusion and the degree hypothesis,

$$\big|N(u) \cap N(v)\big| \;=\; |N(u)| + |N(v)| - \big|N(u) \cup N(v)\big| \;\geq\; \frac{n}{2} + \frac{n}{2} - (n-2) \;=\; 2 \;>\; 0.$$

So there exists (in fact there exist at least two) $w \in N(u) \cap N(v)$, and $u\,w\,v$ is a u–v path of length 2 (it is a path because $w \notin \{u, v\}$).

In both cases $\operatorname{dist}(u, v) \leq 2 < \infty$. As u, v were arbitrary, every pair of vertices is joined by a path, i.e. G is connected, and $\operatorname{diam}(G) \leq 2$.

□

> **ANSWER**
>
> **G is connected.** If it were not, its smallest component H would satisfy $|V(H)| \leq n/2$, and any vertex v of H would have all its neighbours inside H, giving $d(v) \leq |V(H)| - 1 \leq n/2 - 1 < n/2$ — contradicting the hypothesis. The second proof shows more: for non-adjacent u, v the sets $N(u), N(v)$ both live in the $(n-2)$-element set $V \setminus \{u, v\}$ while $|N(u)| + |N(v)| \geq n$, so they must meet; any common neighbour w gives a path $u\,w\,v$. Hence in fact $\operatorname{diam}(G) \leq 2$.

The hypothesis is sharp

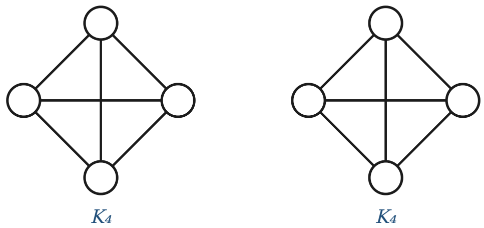
The constant $n/2$ cannot be lowered. For even n, let

$$G_0 \;=\; K_{n/2} \,\dot\cup\, K_{n/2}$$

be the disjoint union of two complete graphs of $n/2$ vertices each. Every vertex of G_0 is adjacent to all the other $n/2 - 1$ vertices of its own clique and to nothing else, so

$$d(v) \;=\; \frac{n}{2} - 1 \qquad \text{for every } v,$$

which is only 1 less than required — and G_0 is disconnected. So the implication genuinely fails as soon as the bound is weakened by one.



$K_4 \mathbin{\dot\cup} K_4$: with $n = 8$, every degree equals $3 = \frac{n}{2} - 1$, yet the graph is disconnected. The threshold in Theorem 7.1 is therefore best possible.

The exact threshold Degrees are integers, so for *odd* n the hypothesis $d(v) \geq n/2$ really says $d(v) \geq \lceil n/2 \rceil = \frac{n+1}{2}$, which is more than necessary. Repeating Proof 1 for odd n: a smallest component satisfies $2h \leq n$, hence $h \leq \frac{n-1}{2}$ and $d(v) \leq \frac{n-3}{2}$; so already $\delta(G) \geq \frac{n-1}{2} = \lfloor n/2 \rfloor$ forces connectivity. Combining the two parities, the sharp statement is:

$$\delta(G) \;\geq\; \left\lfloor \frac{n}{2} \right\rfloor \;\Longrightarrow\; G \text{ is connected,}$$

and this is best possible for every n: the graph $K_{\lfloor n/2 \rfloor} \mathbin{\dot\cup} K_{\lceil n/2 \rceil}$ has $\delta = \lfloor n/2 \rfloor - 1$ and is disconnected.

Context: Dirac's theorem Theorem 7.1 is the easy shadow of a much deeper result with the same hypothesis. *Dirac's theorem* (1952) states that if $n \geq 3$ and $\delta(G) \geq n/2$ then G is *Hamiltonian*: it contains a cycle through all n vertices. A Hamiltonian graph is obviously connected, so Dirac's theorem implies Problem 7 — but "connected" is the part one can prove in three lines, as above. It is instructive that the same threshold $n/2$ is sharp for both statements, and for the same reason: two disjoint cliques.

PROBLEM 8 A directed degree count

directed handshake lemma

PROBLEM

Let us assume that the outdegree of every vertex of a directed graph is 22. It is also known that the indegree of each vertex, except for three special vertices, is 22; furthermore the indegree of two of the three exceptional vertices is 12. What is the indegree of the third exceptional vertex?

Step 1. The directed handshake lemma

Let $D = (V, A)$ be a directed graph: A is a set of *arcs*, each arc a being an ordered pair (x, y) with *tail* x and *head* y. For $v \in V$ write

$$d^+(v) \;=\; \#\{\, a \in A : a \text{ has tail } v \,\} \quad \text{(the }\backslash\text{textit\{outdegree\})}, \qquad d^-(v) \;=\; \#\{\, a \in A : a \text{ has head } v \,\} \quad \text{(the }\backslash\text{textit\{indegree\})}.$$

Theorem 8.1 (directed handshake lemma) *For every finite directed graph $D = (V, A)$,*

$$\sum_{v \in V} d^+(v) \;=\; |A| \;=\; \sum_{v \in V} d^-(v).$$

Proof (double counting). Count the arcs of D by grouping them according to their tail. Every arc has exactly one tail, so the sets $\{a : a \text{ has tail } v\}$, for $v \in V$, partition A; summing their sizes gives $\sum_v d^+(v) = |A|$. Grouping instead by heads: every arc has exactly one head, so the sets $\{a : a \text{ has head } v\}$ also partition A, and $\sum_v d^-(v) = |A|$. □

In words: every arc leaves exactly one vertex and enters exactly one vertex, so the total number of departures, the total number of arrivals, and the number of arcs all coincide. (Unlike the undirected Lemma A there is no factor 2 here: an undirected edge has two *endpoints*, whereas an arc has one tail and one head, counted in two different sums.)

Step 2. Setting up the equation

Let $n = |V|$ be the number of vertices, and denote the three exceptional vertices by w_1, w_2, w_3, where

$$d^-(w_1) = d^-(w_2) = 12, \qquad d^-(w_3) = x \;\; \text{(the unknown).}$$

The data of the problem are then:

- $d^+(v) = 22$ for *every* one of the n vertices;
- $d^-(v) = 22$ for the $n - 3$ vertices other than w_1, w_2, w_3.

Compute $|A|$ twice, using Theorem 8.1. **From the outdegrees:**

$$|A| \;=\; \sum_{v \in V} d^+(v) \;=\; \underbrace{22 + 22 + \cdots + 22}_{n \text{ terms}} \;=\; 22n.$$

From the indegrees, splitting the sum into the ordinary vertices and the three exceptional ones:

$$|A| \;=\; \sum_{v\in V} d^-(v) \;=\; \underbrace{22(n-3)}_{\text{the } n-3 \text{ ordinary vertices}} \;+\; \underbrace{12+12}_{w_1,\,w_2} \;+\; \underbrace{x}_{w_3} \;=\; 22n-66+24+x \;=\; 22n-42+x.$$

Step 3. Solving

Equating the two expressions for the same number $|A|$:

$$22n \;=\; 22n-42+x.$$

The term $22n$ cancels from both sides — note that the answer therefore does not depend on how many vertices the digraph has — leaving

$$0 \;=\; -42+x, \qquad \text{that is} \qquad \boxed{x \;=\; 42}.$$

A useful way to see the same thing without any algebra is as a *balance of deficits and surpluses*. If every vertex had indegree 22, the total indegree would be $22n$, exactly matching the total outdegree. The two vertices with indegree 12 each fall short of 22 by $22-12=10$, a combined deficit of 20. The third exceptional vertex must make up the whole of this deficit on top of its own nominal 22:

$$x \;=\; 22+2\cdot(22-12) \;=\; 22+20 \;=\; 42.$$

ANSWER

The indegree of the third exceptional vertex is 42.

Reason: in any digraph the total outdegree and the total indegree both count the arcs, so they are equal. Here the total outdegree is $22n$, while the total indegree is $22(n-3)+12+12+x = 22n-42+x$. Equating gives $x=42$, independently of the number n of vertices.

Consistency: does such a digraph exist?

The counting argument determines x but does not by itself guarantee that a digraph with these degrees exists; it is worth checking that the answer is not vacuous. If D is required to be a *simple* digraph (no loops, and at most one arc in each direction between any two vertices), then every degree is bounded by $n-1$, so we need

$$d^-(w_3) = 42 \le n-1 \quad\text{and}\quad d^+(v) = 22 \le n-1, \qquad \text{i.e.} \qquad n \;\ge\; 43.$$

And $n \ge 43$ is not merely necessary but sufficient: realising the prescribed pairs of degrees is a bipartite degree-sequence (transportation) problem, and solving it as a max-flow instance produces explicit digraphs. For $n=43$ the resulting digraph has

$$|A| \;=\; 22\cdot 43 \;=\; 946 \text{ arcs}, \qquad d^+ \equiv 22, \qquad d^- \;=\; \big(\underbrace{22,\dots,22}_{40},\, 12,\, 12,\, 42\big),$$

with no loops — and analogous digraphs were constructed for $n=44$ and $n=60$. So the situation described in the problem is genuinely realisable, and 42 is the answer for every admissible size. (If loops or parallel arcs are permitted, the bound $n \ge 43$ disappears and only the counting constraint remains.)

Sanity checks on the answer Two small consistency observations. First, $42 \ne 22$, so w_3 really is "exceptional", as the problem asserts — had the arithmetic produced 22 the hypotheses would have been contradictory. Second, the average indegree must equal the average outdegree, namely 22; and indeed the three exceptional values average out correctly, since $12+12+42 = 66 = 3\cdot 22$.

APPENDIX The graph theory used, problem by problem

a reader's index

The eight problems look varied, but each is settled the same way: identify the right graph, then apply a small number of standard results to it. This appendix records that correspondence, so that it can be checked at a glance.

#	the graph-theoretic objects the solution works with	the results applied to them
1	the knight's graph of the 3×3 board — an isolated vertex plus a single C_8; the four knights as tokens on its vertices	connected 2-regular ⇒ cycle (Thm 3.1); an invariant of the move relation
2	degree sequences; the realising graph G^*	handshake lemma (Lem A); Havel–Hakimi; Erdős–Gallai
3	regularity, Δ, δ; components, trees, paths, cycles	δ ≥ 2 ⇒ a cycle exists (Lem C); degree-closed components (Lem D); a tree has $\lvert V\rvert - 1$ edges
4	vertex–edge incidences; the ray, an infinite graph	double counting (Thm 4.1); reduction mod 2 (Cor 4.2)
5	components; walks and paths; parity of degrees	every walk contains a path (Lem B); Lem D; Cor 4.2 applied to one component
6	the three drawings as circulant graphs $C_9(1,2)$, $C_9(1,3)$, $C_9(1,4)$ on $\mathbb{Z}_9$; complement; Aut	subgraph counts are isomorphism invariants; the multiplier map $x \mapsto ux$ for a unit u
7	components, neighbourhoods, connectivity, diameter	Lem D; inclusion–exclusion on $N(u)$ and $N(v)$; Dirac's theorem for context
8	a digraph: arcs, indegree, outdegree	directed handshake lemma (Thm 8.1); realisation of prescribed degree pairs by max-flow

Table A.1 — what each solution is, in graph-theoretic terms.

Three techniques carry most of the weight

Read across the table and the same three ideas recur; recognising which one a problem calls for is most of the work.

1. Double counting a set of incidences

Count one finite set in two ways and equate the answers. Counting the vertex–edge incidences of a graph by edges and then by vertices gives the handshake lemma $\sum_v d(v) = 2|E|$ (Theorem 4.1); counting the arcs of a digraph by tails and then by heads gives its directed analogue $\sum_v d^+(v) = |A| = \sum_v d^-(v)$ (Theorem 8.1). Everything in Problems 4 and 8 is an immediate consequence, part (b) of Problem 2 is the handshake lemma read modulo 2, Problem 5 is the same statement applied to a single component, and even the classification in Problem 3 has a two-line proof by this route ($2|V(H)| = \sum_{v\in H} d(v) = 2|E(H)|$ forces $|E(H)| = |V(H)|$).

2. Passing to a connected component

A component contains every neighbour of each of its vertices, so degrees measured inside it agree with degrees measured in the whole graph (Lemma D). That innocuous remark is what lets a global counting theorem be applied locally, and it is the pivot of three different arguments: the classification of 2-regular graphs and of graphs with $\Delta \le 2$ (Problem 3, where each component is handled separately), the existence of a second odd-degree vertex *within reach* of the first (Problem 5, where applying the parity corollary to G instead of to the component would prove nothing), and the degree bound $d(v) \le |V(H)| - 1$ inside a smallest component (Problem 7).

3. Exhibiting an invariant

To prove that something *cannot* be done, find a quantity that the permitted operations leave unchanged and show it differs at the two ends. Problems 1 and 6 are the same proof in different clothing:

	Problem 1	Problem 6
what may be done	slide one knight along an edge of C_8 to an empty vertex	relabel the vertices by any bijection
the invariant	the cyclic word of the four knights' colours	the number of triangles
value at the start	W D W D	9 (in G_1)
value at the target	W W D D	3 (in G_2)
conclusion	the target position is unreachable	G_1 and G_2 are not isomorphic

Table A.2 — two impossibility proofs of identical shape: a quantity that the permitted operations cannot change, taking different values at the two ends.

Both invariants are well chosen. In Problem 1 the cyclic colour word is *complete*: Table 1.3 shows that all 140 configurations with the correct pattern really are reachable, so nothing weaker would have sufficed and nothing stronger is true. In Problem 6 the cheap invariants — order, size, degree sequence, connectivity, girth — all agree (Table 6.3), so the triangle count is the first one that does any work.

A closing remark The two halves of an impossibility proof behave quite differently. Showing something *is* possible needs one example, and Problems 2(a) and 6 supply one each. Showing something is *impossible* needs an argument covering every conceivable attempt at once — and an invariant is exactly that, compressed into a single number. Hence Problems 1, 2(b) and 6 all end the same way: a quantity computed twice, giving two different answers.